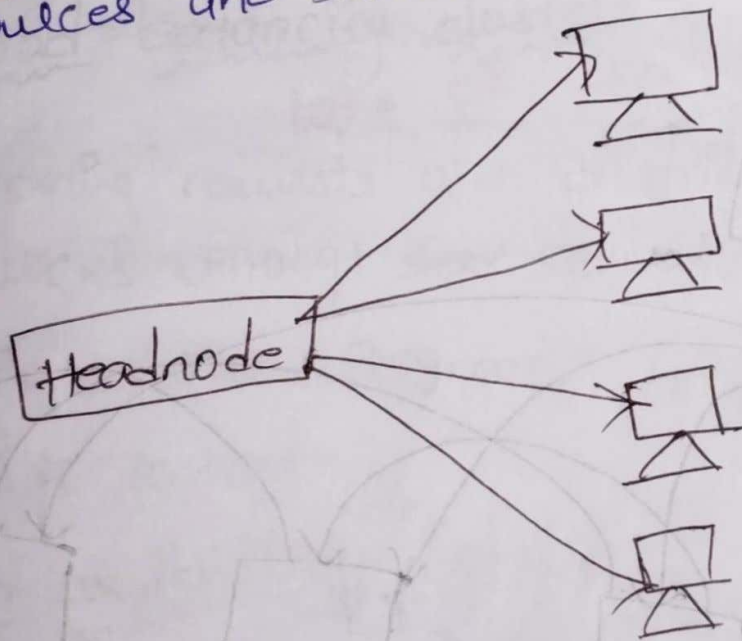


cluster computing:- Multiple computer work together in a single system

cluster computing is the foundation for cloud computing, where multiple interconnected computers work together as a single system to enhance performance, ~~scal~~ scalability and reliability.

* In cloud environment, clusters of virtual machines are used to provide resources and services to users.

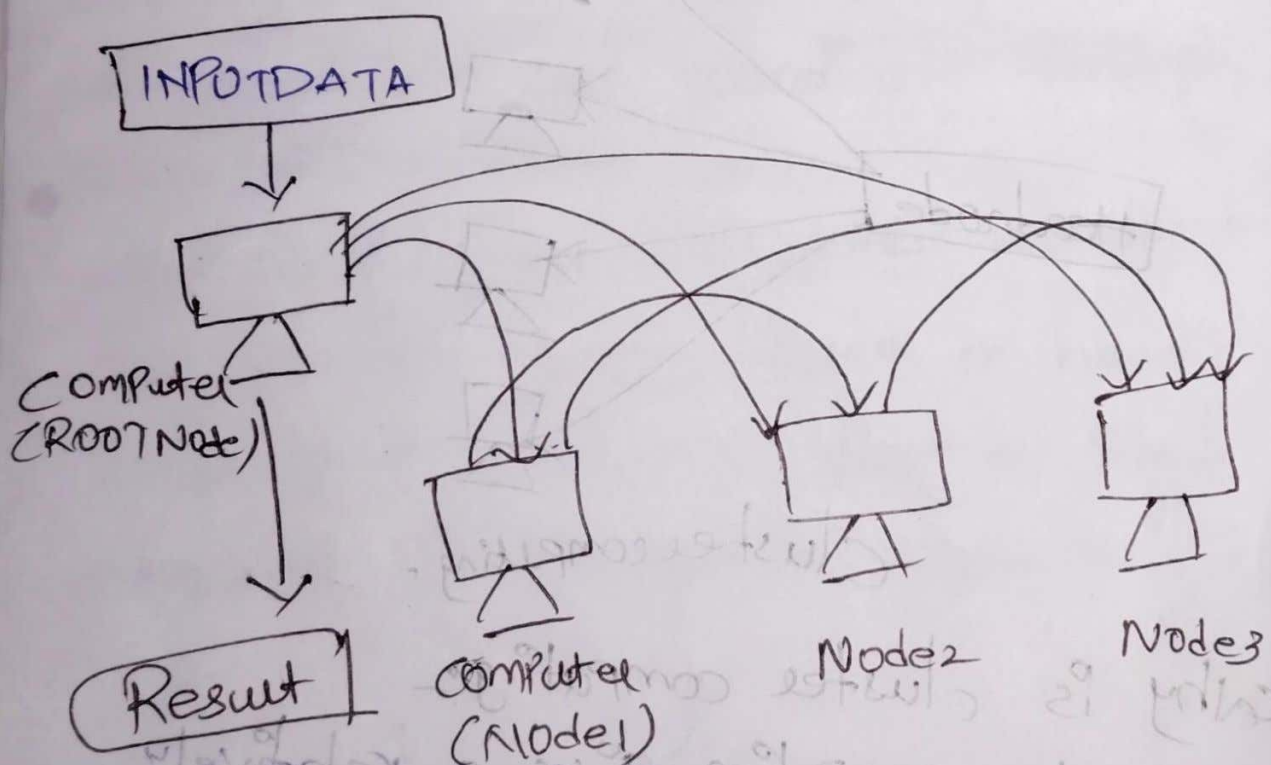


why is cluster computing:-

* cluster computing gives a relatively

in expensive unconventional to the large server or mainframe computer solution.

- * It resolves the demand for content criticality and Process services in a fast way
- * Many ~~organsi~~ organizations and IT companies are implementing cluster computing to augment their scalability, ~~avati~~ availability, processing speed and resource management at Economic prices
- * It ensures that computational power is always available.



Types of cluster computing:-

① HP clusters:- For tasks that need lots of computing power fast.

HP clusters use computer clusters and supercomputers to ~~solve~~ solve advance computational problems. They are used to performing functions that need nodes to communicate as they perform their jobs.

② Load-balancing clusters:- To handle lots of user requests by spreading the work.
incoming requests are distributed for resources among several nodes running similar programs or having similar content.

③ High availability cluster:-

HA cluster are designed to maintain redundant nodes that can act as backup system in case any failure occurs.

* consistent computing services like business activities, complicated database, customer services like e-websites and network file distribution are provided.

classification of cluster:-

1) open cluster:-

~~PA~~ ^{IPs} are needed by every node and ^{corruption prevention} those are accessed only through the internet or web. This type of cluster causes enhanced security concerns.

2) close cluster:-

The nodes are hidden behind the gateway node, and they provide increased protection. They need fewer IP addresses and are good for computational tasks.

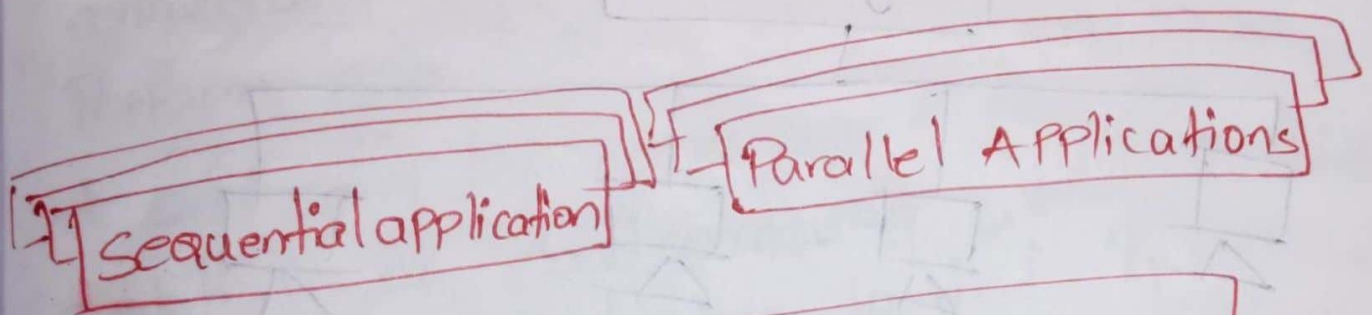
cluster computing architecture:-

* It is designed with an array of interconnected individual computer and the computer systems operating collectively as a single standalone system.

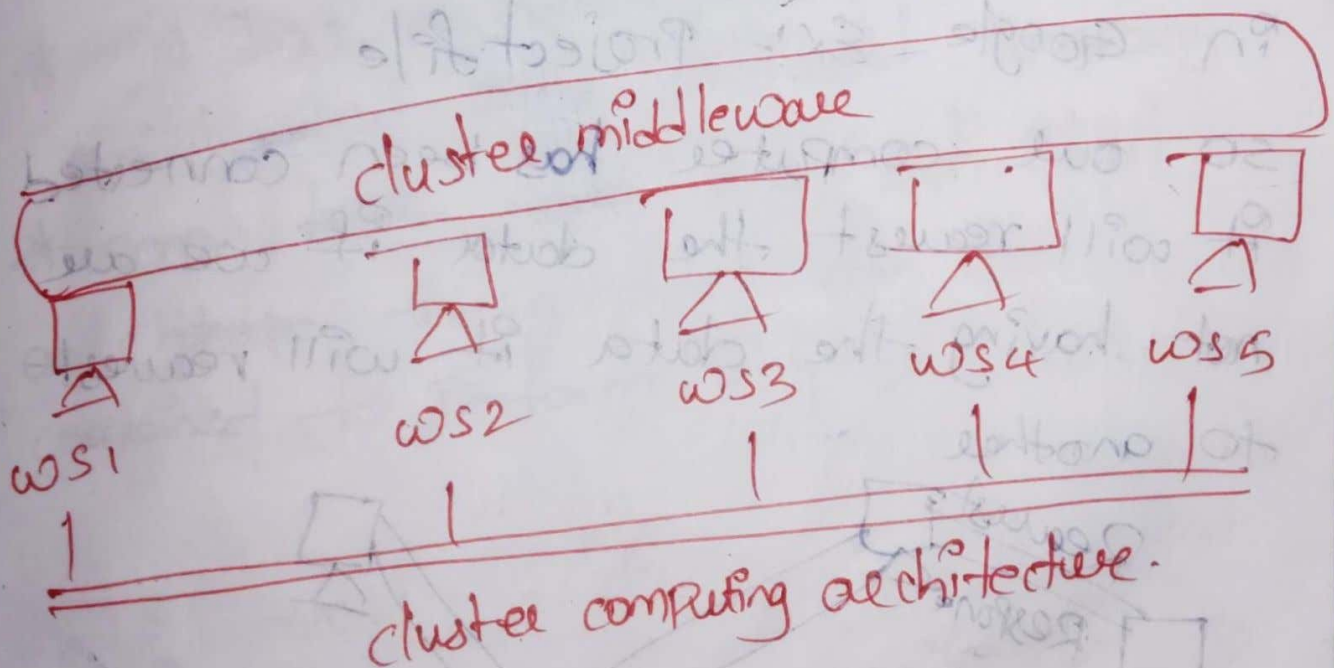
* It is group of workstation or computers working together as a single, integrated computing resource ~~connected~~ connected via high speed interconnect.

* A node - either a single or a multiprocessor network having memory, input and output functions and an operating system.

* Two or more nodes are connected on a single line or every node might be connected individually through a LAN connection



Parallel Programming Environments



Grid computing

Grid computing is a distributed computing model that connects multiple computers or resources across different locations to work together as a single virtual supercomputer to solve large, complex tasks.

* It uses the combined processing power, storage, and data of many computers to perform tasks that require high performance.

Working of Grid computing:-

① Control nodes:-

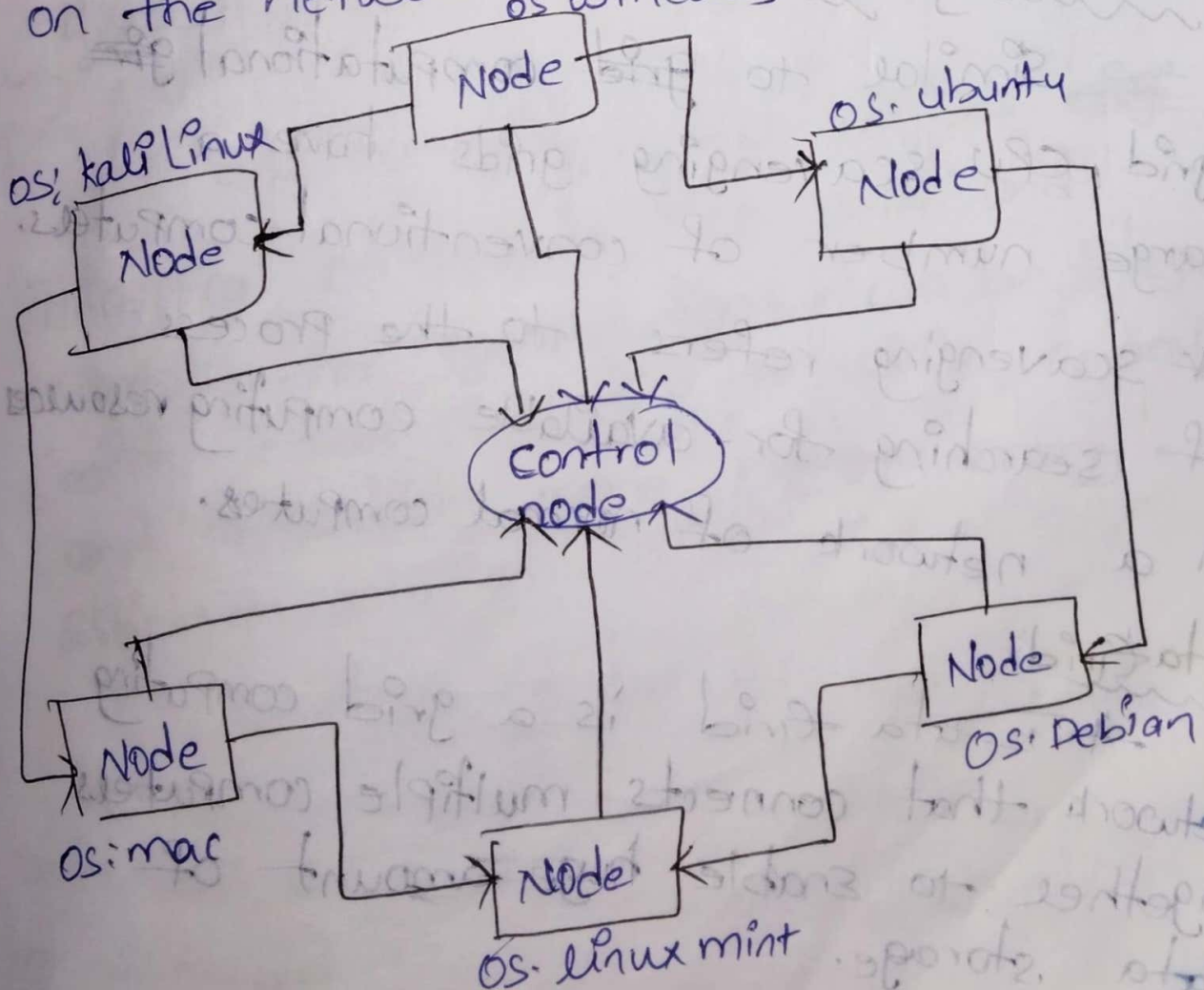
A computer usually a server or a group of servers which administrates the whole network and keeps the account of the resources in the network pool.

② Provider:-

The computer contributes its resources to the network resource pool.

③ User:-

The computer that uses the resources on the network.



Types of Grid computing:-

① Computational Grid

A computational grid is a collection of high-performance processors. It enables researchers to ~~at~~ utilize the combined computing capacity of machines.

* Researchers employ computational grid computing to complete resource-intensive activities like mathematical calculations.

② Scavenging Grid

Similar to ~~grid~~ computational ~~grid~~ grid, CPU scavenging grids have a large number of conventional computers.

* Scavenging refers to the process of searching for available computing resources in a network of normal computers.

Data Grid

A data grid is a grid computing network that connects multiple computers together to enable huge amount of data storage.

* You can access the stored data as if it were on your local system, without worrying about where it is physically located on grid.

Advantages of Grid computing:-

① Grid computing Provide high resources utilization.

② Grid computing allow Parallel Processing of task.

③ Grid computing is designed to be scalable.

BIO-computing:-

* BIO computing, which utilizes ~~biog~~

Biological components like DNA and proteins for computations, can be integrated with cloud computing to offer powerful and potentially more ~~energy~~ energy-efficient solution.

① Data storage and management:-

⇒ cloud Platforms offer scalable and secure storage for vast biological datasets, Protein structures, and experimental data.

② computational Power:-

cloud resources can be dynamically allocated to handle computationally intensive tasks like genome sequencing, Protein structure prediction, and drug discovery.

③ Applications in Drug Discovery:-

Bio computing can be used to simulate biological pathways, identify potential drug targets, and design new therapeutic molecules.

④ cloud-based biocomputing platforms

cloud platforms can host and manage bio computing experiments, providing researchers with access to specialized h/w and s/w tools

What is mobile computing? (MCC)

* Mobile computing is a technology that allows people to access and process data, communicate, and run applications anytime and anywhere, using wireless devices like smartphones, tablets, laptops and wearable devices.

Key features:-

① Portability:-

Devices are lightweight and easy to carry.

② Connectivity:-

uses wireless networks (WiFi, cellular, Bluetooth) to stay connected to the internet or other devices.

③ Interactivity:-

supports user interaction through touch screens, voice inputs, sensors, and cameras.

④ Remote access:-

Allows users to access files, emails and cloud applications with internet access.

characteristics of mobile cloud computing

① Data cache:-

In this, the data can be locally cached.

② User Accommodation:-

Scope of accommodating different user requirements in cloud app development is available in mobile cloud computing.

③ Easy access:-

It is easily accessed from desktop or mobile devices alike.

④ Mobile devices:-

④ Mobile software:- apps and operating systems.

⑤ Wireless networks:- 3G, 4G, 5G, Wi-Fi, Bluetooth

⑥ Cloud services:-

store and process data ~~offline~~ online.

⑦ Improved and increased broadband coverage:-

3G and 4G along with WiFi, femtocells, are providing better connectivity for mobile devices cloud computing.

⑧ Enabling technologies:-

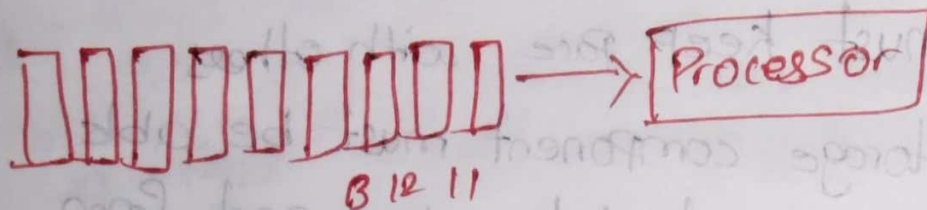
HTML5, CSS3, a hypervisor for mobile devices, cloudlets and web 4.0 are enabling technologies that will drive adoption of mobile cloud computing.

Parallel computing :-

(Contd.)

*

Problem



~~Parallel~~ Parallel computing refers to the

Process of executing several processes on an application or computing simultaneously.

* It is a kind of computing architecture where the large problems break into independent, smaller, usually similar parts.

* It is done by multiple ~~access~~ CPUs continuously communicating via shared memory, which combines results upon completion.

* It helps in performing large computations where the large problem is broken down into more than 1 processor.

* Parallel computing also helps in faster application processing & task resolution by increasing computation power of system.

Reasons:-

- save time & money
- provide concurrency
- solve large problems
- reduce complexity

Where it's used

- ① super computer HPC
- ② weather forecasting
- ③ scientific simulations
- ④ Big data processing (Hadoop)

Ex:- There are 2 persons standing in line for the any ticket.

if 2 cashiers are giving tickets to 2 persons simultaneously, it helps to save time as well as reduce complexity.

Types of Parallel computing:

- ① Bit-level Parallelism
- ② Instruction-level Parallelism
- ③ Task Parallelism.

① Bit-level Parallelism:-

8bit Processor must
two - 16bit in

The form of parallel computing in which every task is dependent on.

In terms of performing a task on large sized data, it reduce the no. of

Instructions the Processor must execute.

* There is need to split the operation into series of instructions.

8bit $\rightarrow$ 16bit - 32bit - 64bit

2.) Instruction level Parallelism:-

* In a single CPU clock cycle the Processor decides in instructions level Parallelism how many instructions are implemented at the same time.

* For each clock cycle phase a Processor in ~~PL~~ instruction level Parallelism can have the ability to address that less than one instruction.

Ex:- super scalar CPUs fetch and execute multiple instruction per clock cycle.

③ Task Parallelism:- Different task on the same or different data.

Task Parallelism is the form of Parallelism in which the tasks are decomposed into subtasks and then allocating each of the subtasks for execution. The processors perform the execution of subtasks concurrently.

Ex:- In a computer - one core runs the UI, another plays audio, another downloads files

